

CAN We Do Better?: Contrasting Security Measures of Controller Area Networks Versus Automotive Ethernet

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I. INTRODUCTION AND MOTIVATION

Modern vehicles are susceptible to attacks through external wireless interfaces like Bluetooth, cellular, or Wi-Fi. If an attacker is able to breach just one gateway within the car, such as the infotainment center, the internal network architecture determines the extent of their access. In a good system, they would not be able to gain access to other systems.

In-Vehicle Networks (IVNs) historically use a Controller Area Network (CAN) bus. Since CAN is a broadcast medium lacking authentication and encryption, one single compromised node can lead to the entire system becoming compromised. In recent years, the automotive industry has started to migrate to using Automotive Ethernet (AE) instead, in order to support high-bandwidth systems like autonomous driving and camera systems.

While manufacturers are typically choosing AE for bandwidth purposes, it also provides a level of security that CAN does not: a switched topology. This projects aims to assess the level of security that AE can provide. Since AE is relatively new technology, and not widely implemented in the majority of cars, CAN still poses a threat to the vehicle security.

Thus, the motivation for this study is to outline the security concerns with CAN and be able to say whether AE is more secure or not. This will pave the way for future research regarding Automotive Ethernet and a new attitude towards it—hopefully one where manufacturers choose AE for more than just bandwidth purposes.

II. BACKGROUND

A. Topology

Most vehicles nowadays are equipped with a CAN bus system that allows different components of cars to talk to each other. For example, the infotainment center may tell you the temperature in the car, your driving speed, or GPS location. All of this information is collected from different nodes in your car. The CAN bus allows them to all communicate with each other.

As seen in Figure 1, the CAN bus terminates at one end and the other is connected to the Electronics Control Unit (ECU). All sensors are connected to both the CAN high and CAN low wires. This allows them to communicate with each other. However, it comes with the drawback that anything connected to the CAN bus can receive all messages sent and even send its own messages to be read by any other sensor attached to the bus. Thus, the CAN bus is insecure by design.

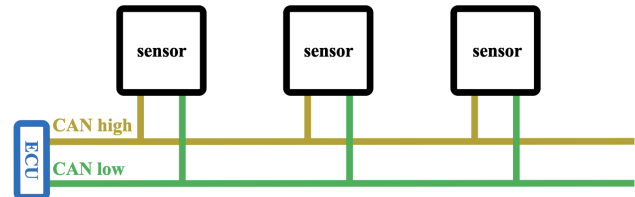


Fig. 1. CAN Bus Topology

In more recent years, manufacturers have been switching to use Automotive Ethernet. They often make this choice since AE is much faster (100 Mb/s) and has much more bandwidth in comparison to CAN (125 Kbit/s)[1], [2]. AE allows for switches to be put in place between the different sensors, which in turn, places a barrier between the sensors' telemetry transmissions.

In Figure 2, the ECU acts as a central switch for the communication between sensors. So, one can not simply connect to any point within the system and read or write messages to any other sensor connected to the system.

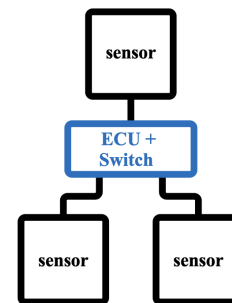


Fig. 2. Automotive Ethernet Topology

Reviewing the topology alone, it is evident that AE is by design, more secure than CAN. So why is AE not universally adopted if it is more secure and faster? A bus topology only requires two wires to connect everything. This cost pennies for manufacturers and is very light. To implement AE, micro-controllers and switches are needed to be able to control the switching mechanism. This is significantly more expensive and heavier than the hardware required for CAN.

TABLE I
RELATED WORKS TABLE

Reference	Contrasts CAN vs AE	Considers Privacy Implications	Assumes a Wireless Compromise	Hardware Validated
Ka et al. (2026)		*	*	
Allen et al. (2024)	*		*	*
Mohammed et al. (2022)			*	*
Won et al. (2026)		*	*	
Won et al. (2025)		*	*	
De Vincenzi et al. (2024)	*			
Oladimeji et al. (2023)		*	*	
Zuo et al. (2021)	*			*
Proposed work	*	*	*	*

B. Protocols

1) *Controller Area Networks*: CAN is defined by the Bosch CAN Specification 2.0 and operates at the physical and data link layers of the the OSI model. Nodes along the bus do not have identifying information, so messages are broadcast with an arbitration identifier. The message with the lowest ID number will be put through first in case of collisions. It also uses two wires to connect all nodes, reducing wiring complexity for the vehicle [2].

Because it is a message based protocol, every node receives a sent message. This could allow for a compromised node to leak all information being sent through the CAN bus. This would include sensitive information like telemetry. Infotainment centers can easily be compromised, allowing for this type of attack to happen. It is also important to note that CAN does not utilize encryption in its message sending. Taking into account these considerations, CAN is inherently insecure.

2) *Automotive Ethernet*: AE is standardized as IEEE 802.3bw-2015 and operates strictly at the physical layer. It reaches speeds of 100 Mbps of bidirectional data transfer over a single unshielded twisted pair of copper wire utilizing PAM3 signaling. Unlike CAN, AE uses a point-to-point connection. Thus, the topology ends up using a single switch for a star or tree topology [1].

The point-to-point nature of AE eliminates the eavesdropping attack vector. A node only sees traffic that the switch allows for it to see. Thus, a compromised node would not be able to see traffic from all other nodes. An attacker would need to compromise the switch itself, which is hidden deep within the hardware of the vehicle, not accessible through wireless connections like the CAN infotainment centers.

III. RESEARCH QUESTIONS

This project seeks to evaluate the security of Controller Area Networks contrasted with Automotive Ethernet by answering two main research questions.

RQ1: Does transitioning from a broadcast CAN bus to a switched Automotive Ethernet network demonstrably eliminate the passive eavesdropping vector from a compromised wireless gateway?

RQ2: What is the extent of data leakage from a compromised infotainment center in both CAN and AE models?

IV. RELATED WORKS

To contextualize the proposed project in relation to existing literature, they are evaluated against four specific criteria pertaining to the research questions outlined.

The first consideration asks whether the literature contrasts Controller Area Networks and Automotive Ethernet. This is the main goal of the proposed project. AE is a relatively new protocol, so the majority of existing literature has yet to consider AE, making it an important thing to note.

Secondly, we assess whether the existing literature considers the privacy implications of the designs. Does the research outlined in the literature measure or address leakage of driver telemetry or other personal information?

We also categorize whether the literature assumes a wireless compromise. Is the threat model based on an attacker pivoting from a remote wireless foothold? This is a popular assumption in the world of vehicle security, as infotainment centers are known for being hackable. So, it is important to note that most literature assumes this fact and uses it for their research.

Lastly, we assess whether the literature validates the hardware through means of a physical test bench or real world vehicle. Researchers, especially those in computer science, often do not test their research on actual or simulated hardware. This is an important step for the proposed research as it will verify whether the attacks can actually occur or not.

In Table I, it is shown that Ka et al. (2026), Won et al. (2026), Won et al. (2025), and Oladimeji et al. (2023) all demonstrate the severe privacy implications of in-vehicle networks when an attacker gains wireless access [3], [4], [5], [6]. These works are some of the best in the field when it comes to addressing the privacy implications, but none of them consider how Automotive Ethernet improves upon the lack of privacy in the CAN model.

While AE is new, there is not much literature analyzing it. However, Zuo et al. (2021) and De Vincenzi et al. (2024) contrast CAN versus AE [7], [8]. Zuo et al. (2021) also backed up their claims with hardware validation, which is not very common among computer science literature regarding CAN and AE networks. Mohammed et al. (2022) demonstrated hardware-validated attacks from a wireless compromise perspective [9].

While this literature is robust and addresses several key points of my research questions, none of them address the four categories outlined for this proposal. However, Allen et al. (2024) represents the closest research to the proposed project. They successfully contrast CAN versus AE under a wireless compromise model and validate their findings using hardware. However, they lack in addressing the privacy concerns of CAN. Their main concern was the safety issues with CAN and they addressed that AE could be a possible solution [10].

This proposed research will improve upon this literature and uniquely address all four points outlined above: comparing CAN versus AE, addressing privacy implications, assuming a wireless compromise, and testing hardware to validate findings.

V. METHODOLOGY

A. Experimental Setup

The methodology utilizes a hybrid approach, combining real-world physical vehicle testing with a network emulation. Future work would be able to expand the emulation into a physical hardware test bench. This test bench will be outlined, but not implemented due to the costliness of the hardware.

The first step will be to connect a passive CAN sniffer to the OBD-II port of my vehicle to collect a live trace of broadcast telemetry during standard operation. The sniffer will be ran on a micro-controller, pulling all the packets from the CAN bus and storing them for inspection. This will represent the data exposed to a compromised gateway on a CAN bus.

The second step will be to do a passive sniffing attempt on an AE equipped vehicle. This will prove that the central gateway blocks passive sniffing of internal node-to-node telemetry. This step exists purely to prove the pre-existing notion of security within Automotive Ethernet, and no data is expected to be pulled beyond any diagnostic codes, as the OBD-II port is designed to only retrieve such codes.

Lastly, to evaluate AE without the constraints of a physical testbench, a software-defined in-vehicle network will be constructed using the Mininet emulator. The simulated topology will mirror a typical setup within an AE equipped vehicle. This will include virtual ECUs, including an engine host, a braking host, and the assumed-compromised infotainment gateway. The real-world CAN telemetry gathered during the physical vehicle tap on the CAN bus equipped vehicles will be translated and replayed across this virtual network using the SOME/IP protocol [11]. By passively monitoring the network interface of the simulated infotainment gateway, the total volume of intercepted telemetry will be measured, and then compared to the CAN telemetry replayed into the system.

All of the data collected in the previous steps will then be analyzed to determine the effectiveness of AE against wireless compromise attacks.

B. Variables

In this setup, the independent variable is the network architecture. These are not going to change or be altered.

they are what we are assessing. The dependent variables will be the privacy leakage obtained. This is represented by the total vehicle telemetry packets successfully intercepted by the compromised nodes. This measured privacy leakage on the CAN bus will be compared against the baseline theoretical vulnerabilities outlined by the related works.

VI. CONCLUSION

This proposed research addresses a critical gap in automotive security and privacy not covered by existing literature. By contrasting the privacy violations in Controller Area Networks and Automotive Ethernet, this research will bridge the gap and define the security of AE in comparison to CAN.

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